Ricardo Emmanuel Uriegas Ibarra

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Software Engineer with 3+ years of experience developing applications and APIs using Python and Java. Proven track record of implementing new features, maintaining backend systems, refactoring code to eliminate bugs and optimize systems performance. Successfully proposed and implemented solutions to optimize storage and processing resources for AI systems.

Experience

Oracle, Software Developer II June 2026 - September 2026 (Caused by Reorg)

Evaluate requirements through Agile to ensure the viability of solution proposals for customers

Design an application for components and LLM/AI workflows that automatized the integration of model fine-tuning, dataset generation, reinforcement learning, A2A Server, A2A Client and inference.

Prioritize dependencies in the development process using Agile to reduce rework and development times.

Validate software functionality through unit and functional testing reducing the risk of undetected implementation discrepancies in the future and producing early corrections.

Update technical documentation for software components, LLM/AI workflows and implementation decisions to ensure technical traceability and decrease software maintenance.

Key Achievements:

Reduced 75% base-model storage requirements in model fine-tuning workflows by the use of LLM/AI models quantization

Planned a 6.2% ROUGE-2 improvement over standard LoRA by integrating AdaLoRA's adaptive parameter allocation into fine-tuning workflows

Planned and implemented a successful re-factorization of nearly 90% of the software's functionalities that reduced the total token consumption per developer

CINVESTAV, Software Developer September 2025 - December 2025

Design a REST API microservices on Docker with PHP and PostgreSQL for a medical platform to avoid rework ensuring findability, accessibility, interoperability and reusability

Develop distributed search engines with API and OWASP HTTPS headers, to optimize catalog retrieval and metadata reuse

Coordinate containerized service deployment with Docker Compose, health checks and persistent logging to ensure availability and traceability of microservices

CINVESTAV, AI Engineer September 2024

Design convolutional neural network (CNN) architecture to optimize the time of discrimination between voice and music classification of audios in a database.

Develop training pipelines for neural networks using TensorFlow/Keras to increment model's precision

Validate model generalization by monitoring performance in Google Colab to prevent accuracy drift.

Key Achievements:

Improved speech and music classification accuracy by 15.8% on a holdout evaluation set of 2,868 audio spectrograms by implementing a Convolutional Neural Network (CNN) with dropout regularization and input normalization.

Reduced model's input size by 99.18% per spectrogram (500×1000 to 64×64 pixels) by integrating an in-model resizing layer into the CNN pipeline

INEEL, Research Intern September 2023

Develop R scripts for causal structure learning with the PC, FCI and RFCI algorithms (pcalg) to ensure reproducible generation of causal graphs (CPDAGs) from tabular data

Design data preprocessing pipelines (categorical encoding and Gaussian transformation) to mitigate violations of conditional independence test assumptions

Key Achievements:

Eliminated memory constraints in PC algorithm execution over 49,019 records by replacing the discrete independence test with a Gaussian test through bestNormalize transformations

Education

B.S. Information Technology Engineering September 2022 - December 2025
Universidad Politécnica de Victoria
Programming Technician July 2019 - August 2022
CBTIS 271 *Mexico*

Papers

Free Alternative for Opinion Mining on Twitter: Integration of Nitter, Puppeteer and VADER December 2025
Published here *DGUTyP*
Developed a free and legal sentiment analysis tool for posts on the social network X, without relying on its official API. The proposed tool uses web scraping via Nitter to collect content, filter by language, and perform sentiment classification using the VADER model. Results are visualized through charts and automated clustering. In tests with the term "Liquid Glass," the system demonstrated acceptable accuracy, achieving an overall F1-score above 81%. This approach offers a cost-free, replicable alternative for researchers or organizations facing economic barriers to accessing social data.

Technical Skills

AI & Machine Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI Systems, LLM Integration, Prompt Engineering, Machine Learning, TensorFlow, PyTorch, AI Prototyping, AI Output Validation

Cloud & DevOps: Microsoft Azure, DevOps, Linux, Bash

Backend & Software Engineering: Python, Java, JavaScript, TypeScript, C/C++, Flask, Django, Spring (Java), FastAPI, Node.js, Microservices, API Development, OOP, Design Patterns, Full-Stack Development, Software Prototyping, React

Data & Databases: SQL, PL/SQL, MySQL, Oracle SQL, Database Design, Data Processing, Data Analysis

Development Tools & Practices: Git, GitHub Copilot, Codex, Claude Code, Cursor, AI-Assisted Development, Agile Methodology, Iterative Development, Automated Testing, Software Quality, Maintainability, Cross-Functional Collaboration, Business-to-AI Use Case Translation

Academic Projects

SQL Interpreter, IDE and Modeler from Scratch
Interpreters *Java, Maven*
- Oriented to csv files management
- Proficient in modeling, generating, and validating SQL scripts for execution across multiple CSV based data repositories functioning as databases.

Stone Detection for Beans
Image Processing *Python, OpenCV, PyQt6*
- Built an algorithm that can detect stones on a photo of beans
- Implemented entirely through traditional image processing, avoiding the need of heavy computational tools

Rotating Tree Fruit Detector Model
Machine Learning *Python, PyQt6, YOLO*
- Live fruit count detector model for ripe berries
- Trained with a short dataset and an accuracy of 87%

Certifications

- Google UX Design 2025
- CCNAv7: Enterprise Networking, Security, and Automation 2024
- Google Project Management 2023
- More if you click here